

When Marginal Shift Magnitudes Cannot Identify Conformal Coverage Loss

CHOROK LEE / KAIST

Equal drift and unchanged confidence can conceal different coverage.

RESEARCH QUESTION

Which distribution summaries identify target prediction-set coverage, and what remains uncertain when the conditional label law is unknown?

40 pp

Sharp limiting gap at $\alpha = 0.1$ in the Gaussian construction

0-20%

Possible errors with unchanged 90% confidence in the $\Gamma = 2$ example

7,200 / 7,200

Simultaneous band containments in controlled replay

Equal magnitude, different consequence

For a symmetric source, equal-length aligned and orthogonal shifts have identical rotation-invariant source-to-target discrepancies. The orthogonal shift preserves scores and coverage; the aligned shift can change coverage. Raw target observations can still distinguish these laws.

A sharp restricted construction

Under a known binary logistic label law and population calibration, a Gaussian scale family approaches excess miscoverage $1/2 - \alpha$. At $\alpha = 0.1$, that is 40 percentage points. The standardized shift diverges; this is not an undetectable-drift example.

When confidence determines coverage

With the known link, the absolute-score distribution is equivalent to the whole coverage curve. Maximum confidence is a one-to-one transformation of score magnitude. This positive result requires the conditional label law, not merely source calibration.

Beyond the known link

Compressed source confidence calibration, target confidence frequencies, and a supplied density-ratio cap yield a sharp target-error interval over compatible hidden source structures. Two target laws attain its lower and upper endpoints simultaneously at every threshold. A fixed full source kernel can give tighter bounds; hidden error variation within confidence levels explains the remaining ambiguity.

Finite-data evidence

Frozen confidence bins yield simultaneous sensitivity intervals from source labels and unlabeled target scores. Controlled learned-score replay gives 7,200/7,200 band containments, but finite bands remain conservative and only 9/72 conditions improve risk-only population bounds.

EVIDENCE AT A GLANCE

Constructed distribution	Constant confidence	Error
Perfectly calibrated source	90%	10%
Target reweights hidden errors up	90%	20%
Target removes hidden error group	90%	0%

TAKEAWAY

Treat drift detection, exact coverage identification, and sensitivity intervals as separate tasks. State which assumptions make each conclusion possible.

INTERPRETATION & LIMITS

Exact confidence sufficiency uses a known binary link. Partial identification retains covariate shift and a specified density-ratio cap, neither certified by unlabeled confidence. Gaussian and replay examples are controlled constructions, not deployment validation. Finite threshold selection uses a fixed grid, not unrestricted repeated looks.